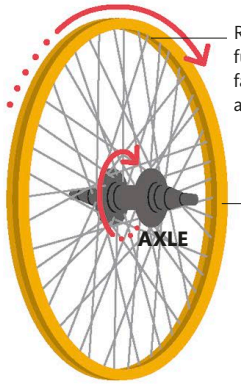
Moving machines

Business, industry, leisure, and tourism rely on fast, long-distance transportation for the movement of goods and people. Transportation technology depends on the use of energy and the application of many different forces to produce motion.

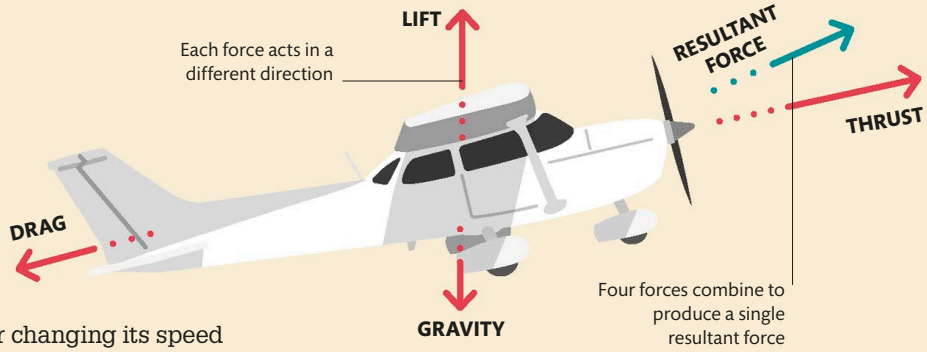
THE WHEEL

The wheel is one of the world's most important inventions. A wheel and axle work like a rotating lever, transmitting force in a circular direction. Turning the wheel around the axle moves the rim of the wheel a greater distance with less force. Turning the rim of the wheel turns the axle with greater force.



Combining forces

An object such as a vehicle moves when it is acted on by one or more forces. As a force is applied, a transfer of energy takes place, either setting the vehicle in motion or changing its speed and direction. Several forces usually act on a vehicle at the same time. Some of the forces may work together, while others work against each other. The combined effect is a single force called the resultant force.



Forces of flight
Four forces are at work when an airplane is in flight. It is pulled down by gravity, pushed upward by lift from its wings, forced forward by thrust from its spinning propeller, and pulled backward by drag. When an airplane is accelerating and climbing, there is an upward resultant force.

Friction

Friction is a force that resists the motion of surfaces sliding across each other. Some friction is necessary; for example, rubber tires rely on friction for grip. However, friction also causes wear and generates heat. Both of these effects are damaging to machines with moving parts. Friction levels depend on the roughness of the surfaces in contact and the amount of force pressing them together. Adding a lubricant reduces friction, because it forms a thin film between the surfaces, keeping them apart.

